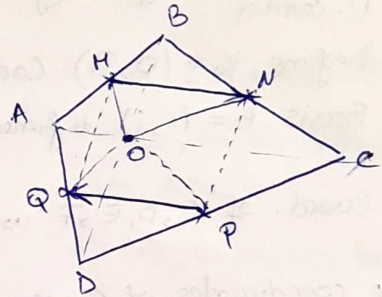


- ① ABCD quadrilateral
M, N, P, Q midpoints

$$\vec{HN} + \vec{PQ} = 0$$

HN PQ parallelogram?



$$\vec{HN} + \vec{PQ} = \vec{HO} + \vec{ON} = \vec{HN}$$

$$\Leftrightarrow -\vec{OH} + \vec{ON} = \vec{HN}$$

$$\Leftrightarrow \vec{ON} - \vec{OH} = \vec{HN}$$

$$\vec{PO} + \vec{OQ} = \vec{PQ}$$

$$\Leftrightarrow \vec{OQ} - \vec{OP} = \vec{PQ}$$

$$\begin{aligned} \vec{HN} + \vec{PQ} &= \vec{ON} - \vec{OH} + \vec{OQ} - \vec{OP} = \frac{\vec{OB} + \vec{OC}}{2} - \frac{\vec{OB} + \vec{OA}}{2} + \frac{\vec{OA} + \vec{OD}}{2} - \frac{\vec{OC} + \vec{OD}}{2} \\ &= \frac{\vec{OB} + \vec{OC} - \vec{OB} - \vec{OA} + \vec{OA} + \vec{OD} - \vec{OC} - \vec{OD}}{2} = 0 \end{aligned}$$

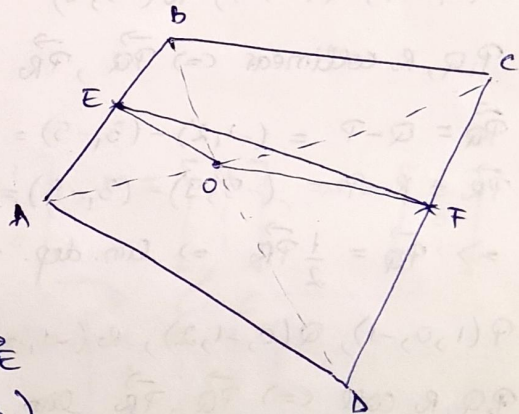
$$\Rightarrow \vec{HN} + \vec{PQ} = 0 \Rightarrow \vec{HN} = -\vec{PQ} \Rightarrow \vec{HN} = \vec{QP} \Rightarrow \text{HN PQ parallelogram.}$$

- ② ABCD quadr.

E - mid. [AB]

F - mid. [CD]

$$\vec{EF} = \frac{1}{2} (\vec{AD} + \vec{BC})$$



$$\vec{EO} + \vec{OF} = \vec{EF}$$

$$\Leftrightarrow -\vec{OE} + \vec{OF} = \vec{EF} \Leftrightarrow \vec{EF} = \vec{OF} - \vec{OE}$$

$$\Leftrightarrow \vec{EF} = \frac{1}{2} (\vec{OC} + \vec{OD}) - \frac{1}{2} (\vec{OA} + \vec{OB})$$

$$= \frac{1}{2} (\vec{OD} + \vec{OC} - \vec{OA} - \vec{OB})$$

$$= \frac{1}{2} (\vec{AD} + \vec{BC})$$

$$\vec{AD} + \vec{OD} = \vec{AD}$$

$$\Rightarrow \vec{AD} = \vec{OD} - \vec{OA}$$

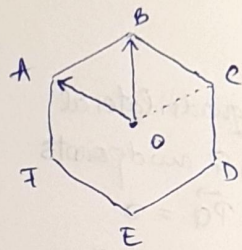
$$\vec{BC} + \vec{OC} = \vec{BC}$$

$$\Rightarrow \vec{BC} = \vec{OC} - \vec{OB}$$

! If ABCD trapezoid, then $\vec{EF}$, $\vec{AD}$, $\vec{BC}$ are collinear and define the same direction

$$|\vec{EF}| = \left| \frac{1}{2} (\vec{AD} + \vec{BC}) \right| = \frac{1}{2} (|\vec{AD}| + |\vec{BC}|)$$

③ plane $\mathbb{E}^2$
 ABCDEF hexagon reg.
 O - center



- Define $K = (O, B)$ Cartesian frame
- Basis $B = (i, j)$ defined by $i = \vec{OA}$
 $j = \vec{OB}$
- Coord. of C, D, E, F w.r. to K.

The coordinates of C are the components of its position vector $\vec{OC}$.

$$\vec{OC} = \vec{OB} + \vec{BC} \xrightarrow{\text{ONBC par.}} \vec{OB} + \vec{AO} = -\vec{OA} + \vec{OB} = -i + j$$

$\Rightarrow C_K (-1, 1) \rightarrow$ cartesian frame of C

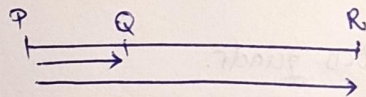
$\vec{OD} = -\vec{OA} = -i \Rightarrow D_K (-1, 0) \rightarrow$ cartesian frame of D

$\vec{OE} = -\vec{OB} = -j \Rightarrow E_K (0, -1) \rightarrow$ cartesian frame of E

$\vec{OF} = -\vec{OC} = i - j \Rightarrow F_K (1, -1) \rightarrow$ cartesian frame of F.

④ Collinear?

a) $P(3, -5), Q(-1, 2), R(-5, 9)$



P, Q, R collinear $\Leftrightarrow \vec{PQ}, \vec{PR}$ lin. dep.

$$\vec{PQ} = Q - P = (-1, 2) - (3, -5) = (-4, 7) \Rightarrow \vec{PQ} (-4, 7)$$

$$\vec{PR} = R - P = (-5, 9) - (3, -5) = (-8, 14) \Rightarrow \vec{PR} (-8, 14)$$

$$\Rightarrow \vec{PQ} = \frac{1}{2} \vec{PR} \Rightarrow \text{lin. dep.} \Rightarrow \text{collinear} \checkmark$$

b) $P(1, 0, -1), Q(0, -1, 2), R(-1, -2, 5)$

P, Q, R coll. $\Leftrightarrow \vec{PQ}, \vec{PR}$ lin. dep.

$$\vec{PQ} (-1, -1, 3) \quad \vec{PR} (-2, -2, 6) \quad \left. \begin{array}{l} \vec{PQ} \\ \vec{PR} \end{array} \right\} \Rightarrow \vec{PQ} = \frac{1}{2} \vec{PR} \Rightarrow \vec{PR} = 2\vec{PQ} \Rightarrow \text{lin. dep.} \Rightarrow \text{collinear} \checkmark$$

$$\vec{PR} (-2, -2, 6)$$

⑤ Form a basis?

a) $v_1(1, 2), v_2(3, 4)$

v_1, v_2 not proportional $\left(\frac{1}{3} \neq \frac{2}{4} = \frac{1}{2} \right) \Rightarrow$ lin. ind.

We see two components $\Rightarrow$ ambient vector space has dim 2

$\Rightarrow (v_1, v_2)$ basis.

b) $v_1(-1, 2, 1, 0)$, $v_2(2, 1, 1, 0)$, $v_3(1, 0, -1, 1)$, $v_4(1, 0, 0, 1)$

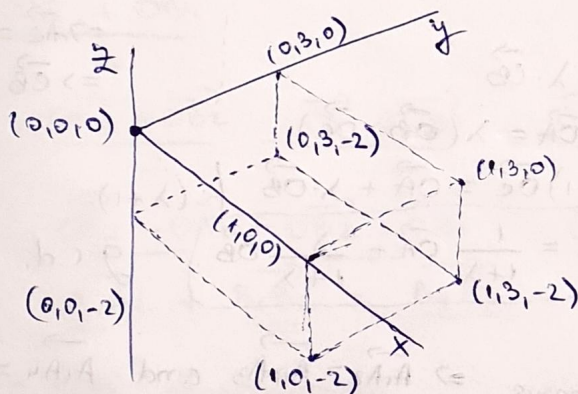
check $\det \neq 0$?

$$\begin{vmatrix} -1 & 2 & 1 & 1 \\ 2 & 1 & 0 & 0 \\ 1 & 1 & -1 & 0 \\ 0 & 0 & 1 & 1 \end{vmatrix} = - \left(\begin{vmatrix} 2 & 1 & 0 \\ 1 & 1 & -1 \\ 0 & 0 & 1 \end{vmatrix} + \begin{vmatrix} -1 & 2 & 1 \\ 2 & 1 & 0 \\ 1 & 1 & -1 \end{vmatrix} \right) = -1 + (1+2-1+4) = 5 \neq 0$$

$v_1 \quad v_2 \quad v_3 \quad v_4$

$\Rightarrow (v_1, v_2, v_3, v_4)$ Basis $\checkmark$.

⑥ Coord. of the vertices of the parallelepiped whose faces lie in the coord. planes and in the planes $x=1$, $y=3$, $z=-2$.



How to draw?

⑦ $B = (i, j, k)$ basis.

$$u = i + j$$

$$v = j + k$$

$$w = i + k$$

$B' = (u, v, w)$? basis

$M_{B'B}, M_{BB'}$?

$$u = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, v = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, w = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$M_{BB'} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \\ \downarrow & \downarrow & \downarrow \\ u & v & w \end{bmatrix}$$

$M_{BB'}$

$$\det M_{BB'} = 2 \neq 0 \Rightarrow \text{basis} \Rightarrow$$

$\Rightarrow B, B'$ have the same orientation

(if B is left oriented then so is B')

$M_{B'B} = M_{BB'}^{-1} \Rightarrow$ calculate OR express i, j, k in terms of u, v, w .

$$u = i + j \Rightarrow i = u - j \Rightarrow \begin{cases} v = j + k \\ w = u - j + k \end{cases}$$

$$\textcircled{+} v + w = u + 2k$$

$$\textcircled{-} v - w = -u + 2j \Rightarrow j = \frac{1}{2}(u + v - w)$$

$$\Rightarrow i = \frac{1}{2}(u - v + w)$$

$$\Rightarrow M_{B'B} = \frac{1}{2} \begin{bmatrix} 1 & 1 & -1 \\ -1 & 1 & 1 \\ 1 & -1 & 1 \\ \downarrow & \downarrow & \downarrow \\ i & j & k \end{bmatrix}$$